

Assignment 4: Binary Message Encoder and Decoder

This program has two parts: a binary message encoder and a binary message decoder. You will submit two separate programs for this assignment on Jupiter Ed: one for the encoder and one for the decoder.

The binary encoder turns a text message into an encoded string of 0s and 1s. The binary decoder turns a string of 0s and 1s into the corresponding text message.

Instructions

The binary encoder should prompt the user for a message, then it should print out the message encoded into binary with a space between each character. Here is some example output:

```
▼ TERMINAL [zsh]
(.venv) $ /Users/sjoiner/src/pyinter-2025/.venv/bin/python /Users/sjoiner/src/pyinter-2025/assignments/assignment-04-encoder-solution.py
Enter a message: Hello!
Your message in binary is: 01001000 01100101 01101100 01101100 01101111 00100001
(.venv) $
```

The binary decoder should prompt the user for a binary encoded message (with a space between each character), then it should print out the decoded message. Here is some example output:

```
▼ TERMINAL [zsh]
(.venv) $ /Users/sjoiner/src/pyinter-2025/.venv/bin/python /Users/sjoiner/src/pyinter-2025/assignments/assignment-04-decoder-solution.py
Enter a binary message: 01001000 01100101 01101100 01101100 01101111 00100001
The message is: Hello!
(.venv) $
```

Hints

1. You don't need any fancy math to do the conversions. You should be able to do them with the functions discussed at the end of the Lecture 4 slides, e.g., `format`, `ord`, `chr`, etc.
2. You might also need some other Python functions such as `input` to get the input message. Also remember your string methods such as `split`.

-
-
3. I recommend getting the encoder working first. Then, you can use its output to test your decoder.

Here is an example of what playing the game could look like.

<pre>Let's play hangman! - - - - - Incorrect guesses: You have 10 guesses left. 135 possible words remain. Guess a letter: a Sorry, a is not in the word. - - - - - Incorrect guesses: a You have 9 guesses left. 72 possible words remain. Guess a letter: u Sorry, u is not in the word. - - - - - Incorrect guesses: a, u You have 8 guesses left. 53 possible words remain. Guess a letter: i Sorry, i is not in the word. - - - - - Incorrect guesses: a, u, i You have 7 guesses left. 16 possible words remain. Guess a letter: o Sorry, o is not in the word. - - - - - Incorrect guesses: a, u, i, o You have 6 guesses left. 9 possible words remain. Guess a letter: e Good guess! e is in the word. _ _ e _ _ e _ _ Incorrect guesses: a, u, i, o You have 6 guesses left. 1 possible word remains. Guess a letter: r Good guess! r is in the word. _ _ e _ _ e r _ Incorrect guesses: a, u, i, o You have 6 guesses left. 1 possible word remains. Guess a letter: s Good guess! s is in the word. s _ e _ _ e r s</pre>		<pre>s _ e _ _ e r s Incorrect guesses: a, u, i, o You have 6 guesses left. 1 possible word remains. Guess a letter: t Good guess! t is in the word. s _ e _ t e r s Incorrect guesses: a, u, i, o You have 6 guesses left. 1 possible word remains. Guess a letter: h Sorry, h is not in the word. s _ e _ t e r s Incorrect guesses: a, u, i, o, h You have 5 guesses left. 1 possible word remains. Guess a letter: l Sorry, l is not in the word. s _ e _ t e r s Incorrect guesses: a, u, i, o, h, l You have 4 guesses left. 1 possible word remains. Guess a letter: c Good guess! c is in the word. s _ e c t e r s Incorrect guesses: a, u, i, o, h, l You have 4 guesses left. 1 possible word remains. Guess a letter: p Good guess! p is in the word. Congratulations! You guessed the word: specters (.venv) \$</pre>
--	--	---